Direct algebraic reductions for the étale argument

What can replace the dynamical rereading?

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The direct flatness implication in this note is now a checked Unit theorem in `sag/etale-direct`. The first part gives mathematical arguments; the last part gives their formalisation correspondence. The residual-cover construction remains a proposal. The two targets must be distinguished: unique square-zero lifting, and a finite cover by *one-variable monic* standard charts.

Part I
Mathematics

1 The lesson from the direct Horrocks argument

The useful simplification in the division proof of rank-one Horrocks was to replace residue-field reasoning with identities over the original ring. An annihilator $\mu \equiv 1 \bmod J$ kills a finite module. The two explicit opens at μ and $1-\mu$ give either the answer or a strict decrease in the polynomial degree. Thus the branching is bounded by an algebraic measure, rather than being obtained by interpreting an arbitrary proof about a generic prime.

The analogous questions here are: which uses of a prime merely select among finitely many available certificates; which residue-field arguments can be replaced by equations over the original ring; and where would a termination measure for a direct covering argument come from?

2 Removing the selected prime from the residual construction

Suppose first that the base is residually discrete local, with residue field k, and that the completed field theorem supplies

$$C = E/\mathfrak{m}E \simeq \prod_i k[T]/(P_i),$$

with P_i monic and separable. Zero factors may be retained. Write e_i for the product idempotents and x_i for generators of the corners, whose identity elements are e_i.

Instead of selecting the factor outside a prime and then selecting x_i or $x_i - 1$, retain both elements

$$y_{i,0} = e_i x_i, \qquad y_{i,1} = e_i(x_i - 1).$$

Here x_i may be any lift to C of the generator of the corner. Then

$$y_{i,0} - y_{i,1} = e_i, \qquad \sum_i (y_{i,0} - y_{i,1}) = 1.$$

This is an actual finite cover certificate. No prime membership test is needed. The translated generator has translated monic separable polynomial, so it carries the same kind of input to the lifting argument.

The useful invariant for each subsequent denominator $h_{i,j}$ is not "outside the selected prime", but an explicit relation

$$y_{i,j}^{N_{i,j}} \in (\overline{h}_{i,j}) \subseteq C.$$

For example, a determinant congruence $p \equiv y^r \bmod \mathfrak{m}$ supplies such information directly. An inverse to a derivative after inverting y supplies it after clearing a power of y. These calculations replace uses of primality with polynomial and localisation identities.

Lemma 1. *If finitely many y_j generate the unit ideal and $y_j^{N_j} \in (h_j)$ with $N_j \geq 1$, then the h_j generate the unit ideal.*

Proof. Raise an identity $1 = \sum_j a_j y_j$ to the power $1 + \sum_j (N_j - 1)$. Every resulting monomial contains some $y_j^{N_j}$, and substitute its given expression as a multiple of h_j. The empty-family case is the zero ring. □

This suggests a precise replacement for the selected-prime local interface: construct finitely many standard charts $E[1/h_j]$, together with an identity

$$1 = \sum_j a_j h_j + \sum_\ell m_\ell b_\ell, \qquad m_\ell \in \mathfrak{m}. \tag{1}$$

The proposal is to obtain this from the existing product theorem and the existing candidate construction, using the power invariant above. This complete residual-cover adapter is not yet formalised. The first-unit and polynomial degree-lowering algorithms still use the discrete residue field; retaining all factors does not remove that input.

3 Why a residual cover is not a global cover

Equation (1) gives $h = \sum_j a_j h_j \in 1 + \mathfrak{m}E$. The charts cover $E[1/h]$. It does not follow that h is a unit in E.

For example, take $R = k[t]_{(t)}$ and

$$E = R \times R[1/t], \qquad \mathfrak{m} = (t).$$

Both factors are étale over R, but $E/\mathfrak{m}E \simeq k$. The idempotent $h = (1, 0)$ satisfies

$$1 - h = (0, 1) = t(0, 1/t) \in \mathfrak{m}E.$$

Nevertheless h is not a unit: its open misses the entire second factor. Thus even the full product decomposition of the closed fibre does not give a global cover of E.

Equation (1) does give a cover by the h_j together with the images of the finitely many m_ℓ. Continuing on the latter opens changes the base to $R[1/m_\ell]$. There is presently no degree or rank bound established for recursively repeating this procedure. An unjustified iteration here would simply hide the remaining local–global problem. For finite R-algebras, the Jacobson-radical property of $\mathfrak{m}E$ would remove this particular obstruction; the original E need not be finite over R.

4 A direct Jacobian source over any base

There is a stronger simplification for the implication from flatness and unramifiedness to square-zero lifting. It needs neither residue fields nor a cover.

Let

$$E = R[X_1, \dots, X_n]/(f_1, \dots, f_m), \qquad x_i = X_i \bmod (f_1, \dots, f_m),$$

and assume formal unramifiedness. The already available diagonal certificate gives $\epsilon \in E \otimes_R E$ with

$$\mu(\epsilon) = 1, \qquad \epsilon(1 \otimes x_i - x_i \otimes 1) = 0.$$

Idempotence of ϵ is not needed in the next calculation.

Lemma 2 (Jacobian certificate from the diagonal)**.** *Put $J_{jk} = (\partial f_j/\partial X_k)(x)$. There is a matrix $C = (c_{ij}) \in M_{n\times m}(E)$ with $CJ = I_n$.*

Proof. Regard the tensor algebra as

$$E[Y_1, \dots, Y_n]/(f_1(Y), \dots, f_m(Y)),$$

and choose a polynomial $p(Y)$ representing ϵ. Then $p(x) = 1$. For each i, kernel membership supplies polynomials $a_{ij}(Y)$ such that, in the polynomial ring over E,

$$p(Y)(Y_i - x_i) = \sum_j a_{ij}(Y) f_j(Y).$$

Differentiate with respect to Y_k and evaluate at $Y = x$. Terms containing $Y_i - x_i$ or $f_j(x)$ vanish, giving

$$\delta_{ik} = \sum_j a_{ij}(x) J_{jk}.$$

Take $c_{ij} = a_{ij}(x)$. Only finitely many representatives and kernel witnesses are used. □

Lift the finitely many c_{ij} to polynomials $\widetilde{c}_{ij} \in R[X]$ and put

$$F_i = \sum_j \widetilde{c}_{ij} f_j, \qquad \Delta = \det\left(\frac{\partial F_i}{\partial X_k}\right).$$

Because every f_j vanishes in E, the Jacobian of the F_i becomes $CJ = I_n$ there. In particular $\Delta(x) = 1$.

Proposition 3. *There is a surjective R-algebra map*

$$u : S = \big(R[X_1, \dots, X_n]/(F_1, \dots, F_n)\big)[1/\Delta] \longrightarrow E.$$

The algebra S is finitely presented and formally étale.

Proof. The map sends X_i to x_i and $1/\Delta$ to 1. It is surjective because the x_i generate E.

For formal étaleness, consider an R-algebra map $S \to D/N$, with $N^2 = 0$. Lift the finitely many coordinates to a tuple $a \in D^n$. The determinant of $J_F(a)$ is a unit modulo N, hence a unit in D. Put

$$b = a - J_F(a)^{-1} F(a).$$

The correction belongs to N^n. The exact first-order Taylor identity modulo N^2 gives $F(b) = 0$. Also $\Delta(b)$ is a unit because its image modulo N is unchanged. This defines a lift. If b' is another lift, Taylor's identity gives $J_F(b)(b' - b) = 0$, hence $b' = b$. For $n = 0$, take the empty determinant 1 and $S = R$. □

This construction avoids a cover by Jacobian minors: suitable linear combinations of the original relations already give an invertible square Jacobian everywhere on the image of E. The statement is a multivariable Jacobian presentation. It does not assert a monic one-variable standard presentation. Compare the distinction in Stacks, Sections 10.143 and 10.144.

5 The short ending using flatness

Use the already proved flatness transfer principle: if S/R is finitely presented and formally unramified, then any R-flat R-algebra E receiving a map from S is flat over S.

Lemma 4 (Lifting through a flat quotient)**.** *Let $u : S \twoheadrightarrow E$ be a surjective map of R-algebras. Suppose S/R is formally smooth and E is flat over S. Then E/R is formally smooth.*

Proof. Write $K = \ker(u)$. A flat quotient has the following explicit property: for each $z \in K$ there is $b \in K$ such that $z = zb$. Given a square-zero ideal $N \subseteq D$ and a map $\alpha : E \to D/N$, lift αu to $v : S \to D$ by formal smoothness of S. For $z \in K$, both $v(z)$ and $v(b)$ belong to N, and therefore

$$v(z) = v(z)v(b) = 0.$$

Thus v factors through E, providing a lift of α. No finite generation of K, chosen idempotent generator, or determinant trick is needed at this step. □

Theorem 5. *A finitely presented, flat, formally unramified R-algebra is formally étale, over an arbitrary commutative ring R.*

Proof. Construct $S \twoheadrightarrow E$ by Proposition 3. Flatness transfer gives S-flatness of E. Lemma 4 gives existence of square-zero lifts; unramifiedness gives uniqueness. □

This proves the flatness implication directly, without ZMT, residue fields, primitive elements, prime ideals, or local–global covering. It does not prove that the multivariable algebra S, or its quotient E, has a finite cover by monic one-variable standard algebras. That is the stronger conclusion still under investigation.

Part II
Connection with Unit

6 The checked arbitrary-base theorem

The whole direct proof is formalised in `sag/etale-direct`, including the diagonal-to-Jacobian step, the constructed source, and the lifting endpoint. In `FlatUnramifiedEtale.Over[R,setR]`, the theorem `Presented[T,presentation,unramified,flat].theorem` has precisely these hypotheses: an actual finite presentation of T, formal unramifiedness, and R-flatness. Its conclusion is the existing `FormallyEtale.EtaleT`.

With a merely given finite presentation, `At[...].formally-etale` still returns an actual unique lifting function. The endpoint `At[...].etale` keeps the presentation-containing conclusion under truncation. Uniqueness makes each type of lifts a proposition, permitting elimination of the finite representative choices without any choice axiom.

7 Correspondence with the mathematical proof

1. `FlatQuotientLift.seg` proves Lemma 4, reusing `FlatQuotient` and `EtaleCase.Surjection.Factor`. No finite kernel generators or idempotent generator are introduced.

2. `ExpressionJets.seg` proves directional differentiation, linearity and the first-order Taylor identity for expression syntax. `FreePolynomialJets.seg` defines differentiation on an actual free polynomial algebra by evaluating into the constructed dual-number ring. `MappedJets.seg` proves compatibility with coefficient change.

3. `DiagonalJacobian.seg` proves Lemma 2. It constructs $E[Y]$ and its quotient by the original mapped equations, then maps the existing diagonal tensor certificate into this quotient. The effective quotient gives the finite coefficient witnesses directly. Thus it does not require a separate ring isomorphism identifying two presentations of the tensor product. `PresentationKernel` is used for equation-indexing helpers; its main kernel theorem is unnecessary here.

4. `JacobianTools.seg` supplies gradient and determinant naturality and linear solving by the existing right adjugate. `JacobianLifting.seg` constructs the finite square-system algebra and its localisation, and proves unique square-zero lifting. This works for every number of variables, including zero.

5. `UnramifiedEtaleSource.seg` lifts the finite coefficients of C, constructs the F_i, proves that their Jacobian becomes I in E, and constructs the surjection $S \twoheadrightarrow E$. Its source certificate exists merely; no flatness is used at this stage.

6. `FlatUnramifiedEtale.seg` composes the source theorem with `FlatDiagonalTransfer` and `FlatQuotientLift`, obtaining Theorem 5 over the original arbitrary base.

The source and all the representative choices remain inside truncations. The new development introduces no axioms. It has only the existing univalence, replacement and surjection parameters; there is no ZMT, field structure, local-ring, prime-ideal or decidability parameter. All reused proof files are relative links and no shared proof or checker file was modified.

8 Validation

A fresh checker run checks 5834 declarations with no failures in 2.26 seconds. The eleven owned proof modules contain 557 declaration events, including nineteen open directives. Each new declaration takes at most 0.02 seconds and opens at most 32 distinct definitions. The frozen opening specification covers all 538 named new declarations; every entry reports no additional openings. This includes concrete quotient carrier and operation aliases.

Regressions cover the empty presentation, $X = a$ and redundant equations, and arbitrary principal localisations. Concrete instances use $\mathbb{Z}/4\mathbb{Z}$, including localisation at its nilpotent element 2. A nonsymmetric two-variable Jacobian checks the orientation and the right-adjugate solver. Mutations of the Newton sign, the defining linear combinations and the Jacobian orientation are rejected in isolated copies.

From `unit/`, run `./segc sag/etale-direct all`. The directory's `README.txt` gives the exact exports and audit commands.

9 Assessment for the original global structure theorem

There are now three distinct tasks rather than a single opaque dynamical transformation:

1. The direct flatness implication above is now checked over an arbitrary base, with its source and all intermediate certificates constructed.

2. The prime-free residual-cover adapter removes factor selection and the x versus $x-1$ decision from the local structure proof. It reuses the completed field product theorem, but still has a discrete residue field and only produces a cover modulo $\mathfrak{m}E$.

3. The full standard-cover theorem still needs a direct finite one-variable chart construction over an arbitrary base, or a justified finite local–global argument. Neither the Jacobian presentation nor the residual-cover identity supplies that step. The two chart descent lemmas from the original note remain independently useful.

The analysis therefore supports simplifying the algebra before building a general dynamical framework. It gives a definite shortcut for the flatness implication and a sharper, genuinely smaller target for the remaining global structure problem; it does not claim that all dynamical reasoning has been eliminated from that stronger problem.